Beta-casomorphins-7 in infants on different type of feeding and different levels of psychomotor development.

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Casomorphins are the most important during the first year of life, when postnatal formation is most active and milk is the main source of both nutritive and biologically active material for infants. This study was conducted on a total of 90 infants, of which 37 were fed with breast milk and 53 were fed with formula containing cow milk. The study has firstly indicated substances with immunoreactivity of human (irHCM) and bovine (irBCM) beta-casomorphins-7 in blood plasma of naturally and artificially fed infants, respectively. irHCM and irBCM were detected both in the morning before feeding (basal level), and 3h after feeding. Elevation of irHCM and irBCM levels after feeding was detected mainly in infants in the first 3 months of life. Chromatographic characterization of the material with irBCM has demonstrated that it has the same molecular mass and polarity as synthetic bovine beta-casomorphin-7. The highest basal irHCM was observed in breast-fed infants with normal psychomotor development and muscle tone. In contrast, elevated basal irBCM was found in formula-fed infants showing delay in psychomotor development and heightened muscle tone. Among formula-fed infants with normal development, the rate of this parameter directly correlated to basal irBCM. The data indicate that breast feeding has an advantage over artificial feeding for infants' development during the first year of life and support the hypothesis for deterioration of bovine casomorphin elimination as a risk factor for delay in psychomotor development and other diseases such as autism.